

CS 305 Lab Tutorial

Lab 7 UDP&TCP

Dept. Computer Science and Engineering
Southern University of Science and Technology

Topics

- UDP
 - UDP segment structure
- TCP
 - TCP connection
 - TCP segment structure
 - Sliding window system
 - Retransmission
- Practices

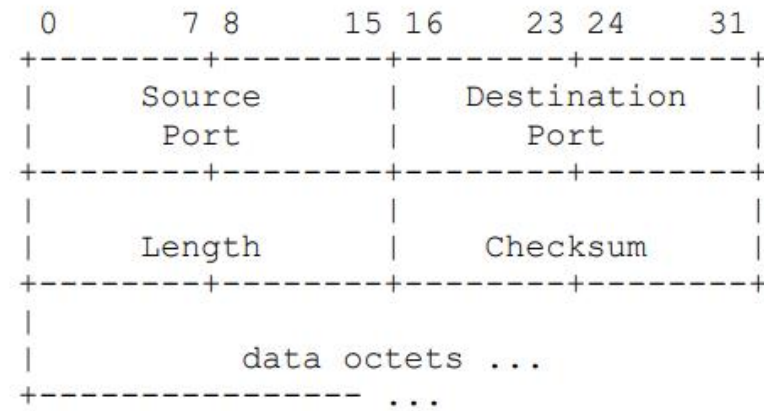
Part A. UDP

- UDP: User Datagram Protocol
- UDP is defined to make available a datagram mode of packet-switched computer communication in the environment of an interconnected set of computer networks.
- UDP assumes that the Internet Protocol (IP) is used as the underlying protocol.
- UDP is transaction oriented, and delivery and duplicate protection are NOT guaranteed.

<https://tools.ietf.org/html/rfc768>

UDP segment structure

- Source port: may be zero or indicates the port of the sending process
- Destination port: an internet destination address
- Length: in octets of this user datagram including this header and the data. (This means the minimum value of the length is eight.)



User Datagram Header Format

<https://tools.ietf.org/html/rfc768>

A UDP segment(1)

No.	Time	Source	Destination	Protocol	Length	Info
2732	17.881663	10.21.3.80	172.18.1.92	DNS	83	Standard query 0x0004 A www.sustc.edu.cn.edu.cn
2733	17.924398	172.18.1.92	10.21.3.80	DNS	83	Standard query response 0x0004 Server failure A

> Frame 2732: 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface 0

> Ethernet II, Src: IntelCor 5c:69:58 (90:61:ae:5c:69:58), Dst: JuniperN_aa:6d:c3 (2c:21:31:aa:6d:c3)

> **Internet Protocol Version 4**, Src: 10.21.3.80, Dst: 172.18.1.92

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 69

Identification: 0x4ca7 (19623)

> Flags: 0x0000

Time to live: 64

Protocol: UDP (17)

Header checksum: 0x732e [validation disabled]

[Header checksum status: Unverified]

Source: 10.21.3.80

Destination: 172.18.1.92

> **User Datagram Protocol**, Src Port: 64176, Dst Port: 53

Source Port: 64176

Destination Port: 53

Length: 49

Checksum: 0xc67f [unverified]

[Checksum Status: Unverified]

[Stream index: 468]

> Domain Name System (query)

UDP segment is the payload of IP package

UDP is identified by 17 in protocol field of IP package

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A UDP segment(2)

No.	Time	Source	Destination	Protocol	Length	Info
2732	17.881663	10.21.3.80	172.18.1.92	DNS	83	Standard query 0x0004 A www.sustc.edu.cn.edu.cn
2733	17.924398	172.18.1.92	10.21.3.80	DNS	83	Standard query response 0x0004 Server failure A

> Frame 2732: 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface 0

> Ethernet II, Src: IntelCor_5c:69:58 (90:61:ae:5c:69:58), Dst: JuniperN_aa:6d:c3 (2c:21:31:aa:6d:c3)

> Internet Protocol Version 4, Src: 10.21.3.80, Dst: 172.18.1.92

> User Datagram Protocol, Src Port: 64176, Dst Port: 53

Source Port: 64176

Destination Port: 53

Length: 49

Checksum: 0xc67f [unverified]

[Checksum Status: Unverified]

[Stream index: 468]

> Domain Name System (query)

Transaction ID: 0x0004

> Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

> Queries

[Response In: 2733]

UDP head, including source port,dest port,length and checksum

here the DNS data is the payload of UDP

0000 2c 21 31 aa 6d c3 90 61 ae 5c 69 58 08 00 45 00 ,!1m..a .\iX..E.
0010 00 45 4c a7 00 00 40 11 73 2e 0a 15 03 50 ac 12 .EL...@. s....P..
0020 01 5c fa b0 00 35 00 31 c6 7f 00 04 01 00 00 01 .\...5.1
0030 00 00 00 00 00 00 03 77 77 77 05 73 75 73 74 63w ww.sustc
0040 03 65 64 75 02 63 6e 03 65 64 75 02 63 6e 00 00 .edu.cn. edu.cn..
0050 01 00 01

While invoke an DNS query, this session is using UDP as transport protocol
You can use 'nslookup' or 'dig' to invoke an DNS query

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Part B. TCP

- TCP: Transmission Control Protocol
- TCP is a highly reliable host-to-host protocol between hosts in packet-switched computer communication networks, and in interconnected systems of such networks.
- TCP must recover from data that is damaged, lost, duplicated, or delivered out of order by the Internet communication system.
 - Ports
 - connections
 - Flow control
 - Reliability

<https://tools.ietf.org/html/rfc793>

Part B.1 A TCP connection(1)

tcp.stream eq 0

No.	Time	Source	Destination	Protoc	Info	
4	0.350305	192.168.88.149	14.215.177.39	TCP	60920 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1	connection establish
5	0.448978	14.215.177.39	192.168.88.149	TCP	80 → 60920 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1452 WS=32 SACK_PERM=1	
6	0.449087	192.168.88.149	14.215.177.39	TCP	60920 → 80 [ACK] Seq=1 Ack=1 Win=66560 Len=0	
7	0.449211	192.168.88.149	14.215.177.39	HTTP	HEAD / HTTP/1.1	http over tcp
8	0.487134	14.215.177.39	192.168.88.149	TCP	80 → 60920 [ACK] Seq=1 Ack=79 Win=24832 Len=0	
9	0.493653	14.215.177.39	192.168.88.149	HTTP	HTTP/1.1 200 OK	
10	0.497383	192.168.88.149	14.215.177.39	TCP	60920 → 80 [FIN, ACK] Seq=79 Ack=333 Win=66304 Len=0	connection close
12	0.563547	14.215.177.39	192.168.88.149	TCP	80 → 60920 [ACK] Seq=333 Ack=80 Win=24832 Len=0	
13	0.566737	14.215.177.39	192.168.88.149	TCP	80 → 60920 [FIN, ACK] Seq=333 Ack=80 Win=24832 Len=0	
14	0.566805	192.168.88.149	14.215.177.39	TCP	60920 → 80 [ACK] Seq=80 Ack=334 Win=66304 Len=0	

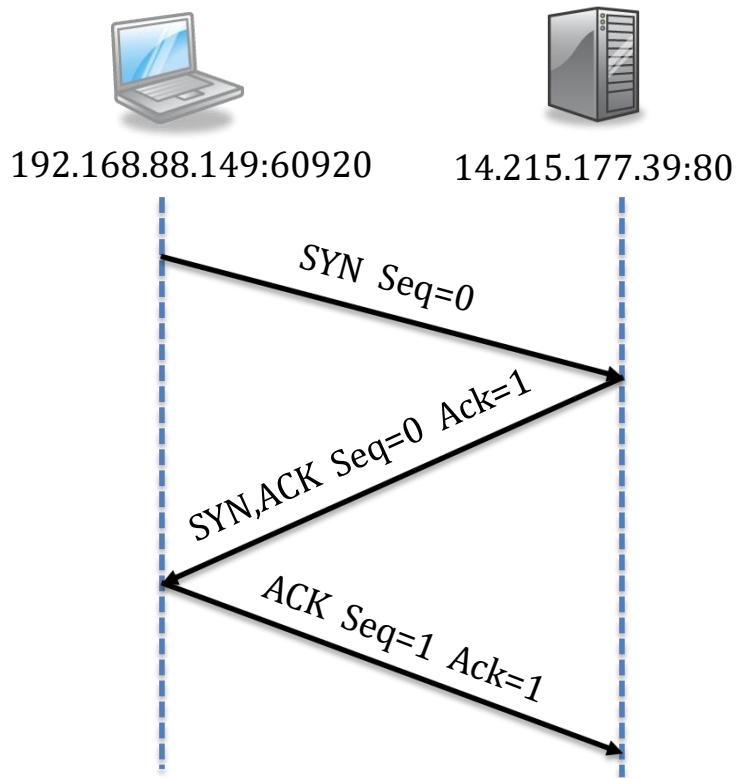
Source	Destination	Protoc	Info
192.168.88.149	14.215.177.39	TCP	60920 → 80

Source IP:192.168.88.149 port: 60920

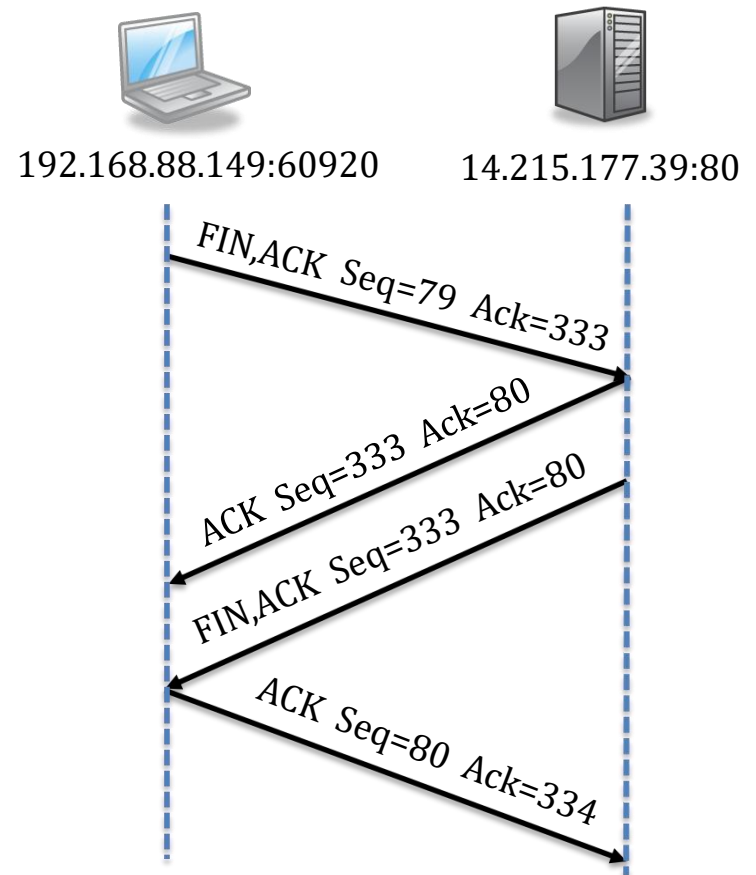
Destination IP:14.215.177.39 port:80

Tips: Using command 'curl' to invoke a http request which using TCP for transport
For example: curl -i www.baidu.com

Part B.1 A TCP connection(2)

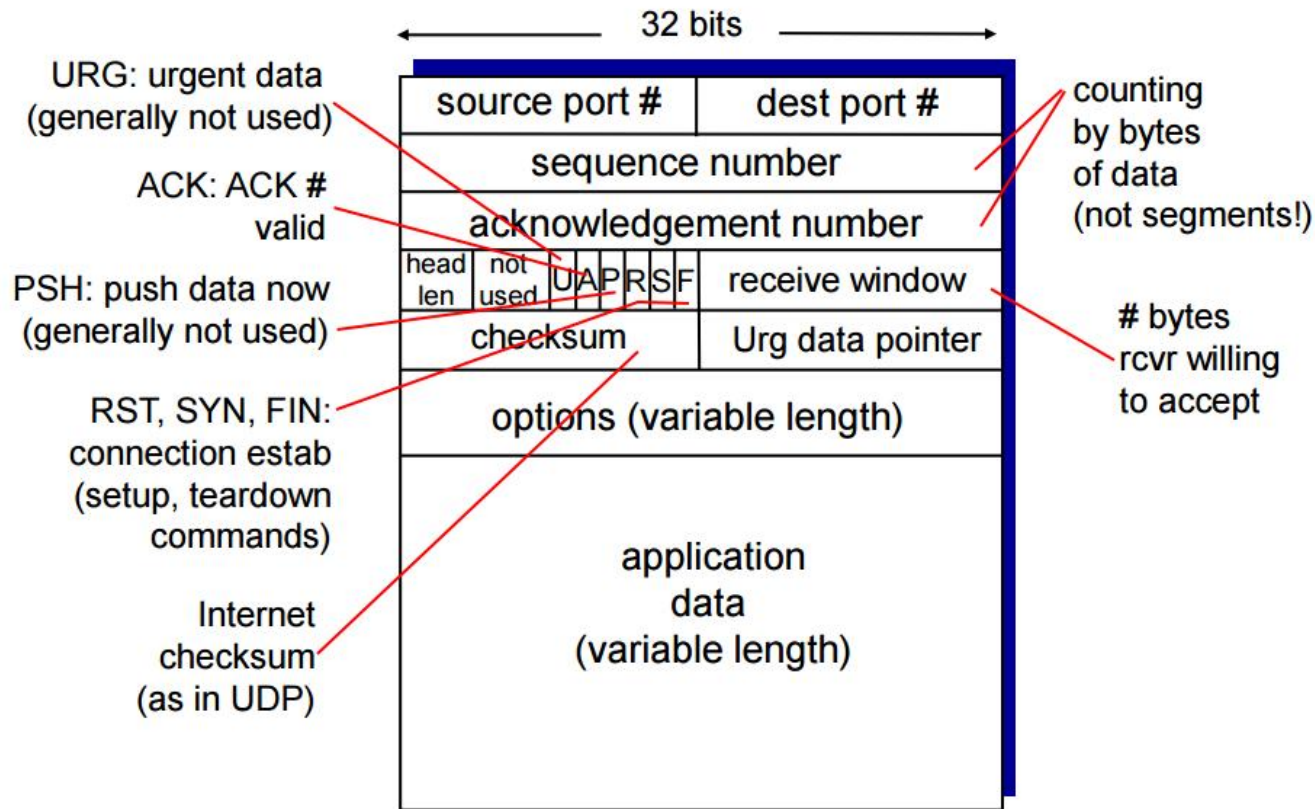


Establishing a TCP connection



Closing a TCP connection

Part B.2 TCP segment structure



'Sequence number' and 'Ack number'(1)

Transmission is made reliable via the use of **sequence numbers** and **acknowledgments**.

- The sequence number **of the first octet of data in a segment** is transmitted with that segment and is called the segment **sequence number**.
- Segments also carry an **acknowledgment number** which is the sequence number of the next expected data octet of transmissions in the reverse direction.

When the TCP transmits a segment containing data, it puts a copy on a retransmission queue and starts a timer;

- when the acknowledgment for that data is received, the segment is deleted from the queue.
- If the acknowledgment is not received before the timer runs out, the segment is retransmitted.

An acknowledgment by TCP does not guarantee that the data has been delivered to the end user, but only that the receiving TCP has taken the responsibility to do so.

<https://tools.ietf.org/html/rfc793>

'Sequence number' and 'Ack number'(2)

```
Transmission Control Protocol, Src Port: 80, Dst Port: 54861, Seq: 81761, Ack: 333, Len: 1460
Source Port: 80
Destination Port: 54861
[Stream index: 2]
[TCP Segment Len: 1460]
Sequence number: 81761 (relative sequence number)
[Next sequence number: 83221 (relative sequence number)]
Acknowledgment number: 333 (relative ack number)
```

No.	Time	Source	Destination	Protoc	Info
234	10.752731	192.168.88.149	128.119.245.12	TCP	54861 → 80 [ACK] Seq=333 Ack=81761 Win=55296 Len=0
235	11.462632	128.119.245.12	192.168.88.149	TCP	80 → 54861 [ACK] Seq=81761 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]
236	11.463266	128.119.245.12	192.168.88.149	TCP	80 → 54861 [ACK] Seq=83221 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]
237	11.463358	192.168.88.149	128.119.245.12	TCP	54861 → 80 [ACK] Seq=333 Ack=84681 Win=52480 Len=0

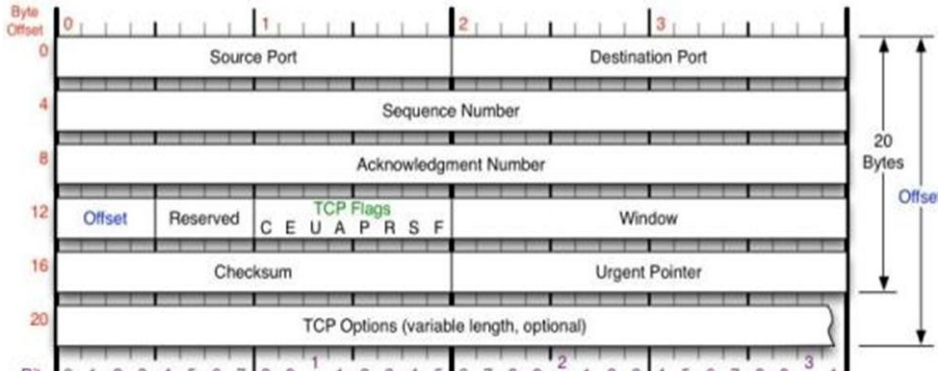
54861->80: seq = 333 len=0

80->54861: ack=333+0 seq = 81761 len=1460

80->54861: ack=333+0 seq = 83221(81761+1460) len=1460

54861->80: seq = 333(333+0) ack=84681(83221+1460) len=0

'Header len'/'offset' field in TCP header



```

> Transmission Control Protocol, Src Port: 54861, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
  Source Port: 54861
  Destination Port: 80
  [Stream index: 2]
  [TCP Segment Len: 0]
  Sequence number: 1 (relative sequence number)
  [Next sequence number: 1 (relative sequence number)]
  Acknowledgment number: 1 (relative ack number)
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x010 (ACK)
  Window size value: 256
  [Calculated window size: 65536]
  [Window size scaling factor: 256]
  Checksum: 0x13ef [unverified]
  [Checksum Status: Unverified]
  Urgent pointer: 0
    
```

head length is 20 byte
while there's no options

Data Offset: 4 bits

- The number of 32 bit words in the TCP Header. This indicates where the data begins.
- The TCP header (even one including options) is an integral number of 32 bits long.

```

> Transmission Control Protocol, Src Port: 54861, Dst Port: 80, Seq: 0, Len: 0
  Source Port: 54861
  Destination Port: 80
  [Stream index: 2]
  [TCP Segment Len: 0]
  Sequence number: 0 (relative sequence number)
  [Next sequence number: 0 (relative sequence number)]
  Acknowledgment number: 0
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x002 (SYN)
  Window size value: 64240
  [Calculated window size: 64240]
  Checksum: 0x5335 [unverified]
  [Checksum Status: Unverified]
  Urgent pointer: 0
  > Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, N
    
```

32 bytes = 8*4bytes

32bytes = 20(default length) + 12
(options length)

'Flags' in TCP header

```
Flags: 0x002 (SYN)
000. .... = Reserved: Not set
...0 .... = Nonce: Not set
.... 0... = Congestion Window Reduced (CWR): Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...0 = Acknowledgment: Not set
.... ....0 = Push: Not set
.... .....0 = Reset: Not set
> ..... 1. = Syn: Set
```

```
Flags: 0x012 (SYN, ACK)
000. .... = Reserved: Not set
...0 .... = Nonce: Not set
.... 0... = Congestion Window Reduced (CWR): Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... ....0 = Push: Not set
.... .....0 = Reset: Not set
> ..... 1. = Syn: Set
.... ....0 = Fin: Not set
```

```
Flags: 0x010 (ACK)
000. .... = Reserved: Not set
...0 .... = Nonce: Not set
.... 0... = Congestion Window Reduced (CWR): Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... ....0 = Push: Not set
.... .....0 = Reset: Not set
.... ....0 = Syn: Not set
.... .....0 = Fin: Not set
[TCP Flags: .....A....]
```

Control Bits:

URG: Urgent Pointer field significant

ACK: Acknowledgment field significant

PSH: Push Function

RST: Reset the connection

SYN: Synchronize sequence numbers

FIN: No more data from sender

```
Flags: 0x011 (FIN, ACK)
000. .... = Reserved: Not set
...0 .... = Nonce: Not set
.... 0... = Congestion Window Reduced (CWR): Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... ....0 = Push: Not set
.... .....0 = Reset: Not set
> ..... 1. = Fin: Set
[TCP Flags: .....A...F]
```

```
Flags: 0x019 (FIN, PSH, ACK)
000. .... = Reserved: Not set
...0 .... = Nonce: Not set
.... 0... = Congestion Window Reduced (CWR): Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... ....1 = Push: Set
.... .....0 = Reset: Not set
.... ....0 = Syn: Not set
> ..... 1. = Fin: Set
```

Tips in Wireshark: Using 'tcp.flags.xxx==1' as filter to view the corresponding package
While xxx is the name of the flag, such as tcp.flags.syn==1

'Window' field in TCP header

- TCP provides a means for the receiver to govern the amount of data sent by the sender. This is achieved by returning a "window" with every ACK indicating a range of acceptable sequence numbers beyond the last segment successfully received.
- The window indicates an allowed number of octets that the sender may transmit before receiving further permission.

tcp.stream eq 2 && tcp.dstport==80									
No.	Time	Source	Destination	Protoc	Info				
296	18.363331	192.168.88.149	128.119.245.12	TCP	54861 → 80	[ACK]	Seq=333	Ack=127021	Win=9984
298	18.405271	192.168.88.149	128.119.245.12	TCP	54861 → 80	[ACK]	Seq=333	Ack=128481	Win=8704
301	18.746754	192.168.88.149	128.119.245.12	TCP	54861 → 80	[ACK]	Seq=333	Ack=131401	Win=5632
303	18.787241	192.168.88.149	128.119.245.12	TCP	54861 → 80	[ACK]	Seq=333	Ack=132861	Win=4352
307	19.117577	192.168.88.149	128.119.245.12	TCP	54861 → 80	[ACK]	Seq=333	Ack=135781	Win=1280

the size of rcv window is dynamic changing

```
Transmission Control Protocol, Src Port: 54861, Dst Port: 80, Seq: 0, Len: 0
Source Port: 54861
Destination Port: 80
[Stream index: 2]
[TCP Segment Len: 0]
Sequence number: 0 (relative sequence number)
[Next sequence number: 0 (relative sequence number)]
Acknowledgment number: 0
1000 .... = Header Length: 32 bytes (8)
> Flags: 0x002 (SYN)
Window size value: 64240
[Calculated window size: 64240]
Checksum: 0x5335 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale,
> TCP Option - Maximum segment size: 1460 bytes
> TCP Option - No-Operation (NOP)
> TCP Option - Window scale: 8 (multiply by 256)
Kind: Window Scale (3)
Length: 3
Shift count: 8
[Multiplier: 256]
```

while in SYN, the multiplier on window is determined by 'window scale option'

```
No. Time Source Destination Protoc Info
296 18.363331 192.168.88.149 128.119.245.12 TCP 54861 → 80 [ACK] Seq=333 Ack=127021 Win=9984 Len=0
298 18.405271 192.168.88.149 128.119.245.12 TCP 54861 → 80 [ACK] Seq=333 Ack=128481 Win=8704 Len=0
301 18.746754 192.168.88.149 128.119.245.12 TCP 54861 → 80 [ACK] Seq=333 Ack=131401 Win=5632 Len=0
303 18.787241 192.168.88.149 128.119.245.12 TCP 54861 → 80 [ACK] Seq=333 Ack=132861 Win=4352 Len=0
307 19.117577 192.168.88.149 128.119.245.12 TCP 54861 → 80 [ACK] Seq=333 Ack=135781 Win=1280 Len=0

> Frame 296: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
> Ethernet II, Src: IntelCor_Sc:69:58 (90:61:ae:5c:69:58), Dst: Routerbo_bd:b8:f5 (00:0c:42:bd:b8:f5)
> Internet Protocol Version 4, Src: 192.168.88.149, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 54861, Dst Port: 80, Seq: 333, Ack: 127021, Len: 0
Source Port: 54861
Destination Port: 80
[Stream index: 2]
[TCP Segment Len: 0]
Sequence number: 333 (relative sequence number)
[Next sequence number: 333 (relative sequence number)]
Acknowledgment number: 127021 (relative ack number)
0101 .... = Header Length: 20 bytes (5)
> Flags: 0x010 (ACK)
Window size value: 39
[Calculated window size: 9984]
Window size scaling factor: 256
Checksum: 0x234e [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0

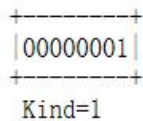
0000 00 0c 42 bd b8 f5 90 61 ae 5c 69 58 08 00 45 00 ..B....a..ix..E.
0010 00 28 53 c1 40 00 40 06 58 4d c0 a8 58 95 80 77 .(S.@: XM.X.w
0020 f5 0c d6 4d 00 50 d1 8b eb 4b 91 50 d8 d7 50 10 ..M.P..K.P.P.
0030 00 27 23 4e 00 00 ..#N..
```

9984 = 39(size value) * 256(scaling factor)

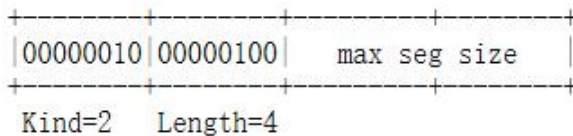
'Options'(variable) in TCP header

- May occupy **space at the end of the TCP header**
- **a multiple of 8 bits in length.**
- **No-operation** may be used between options, for example, to align the beginning of a subsequent option on a word boundary.

No-Operation



Maximum Segment Size



Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Op

✓ TCP Option - Maximum segment size: 1460 bytes

Kind: Maximum Segment Size (2)

Length: 4

MSS Value: 1460

> TCP Option - No-Operation (NOP)

✓ TCP Option - Window scale: 8 (multiply by 256)

Kind: Window Scale (3)

Length: 3

Shift count: 8

[Multiplier: 256]

> TCP Option - No-Operation (NOP)

> TCP Option - No-Operation (NOP)

✓ TCP Option - SACK permitted

Kind: SACK Permitted (4)

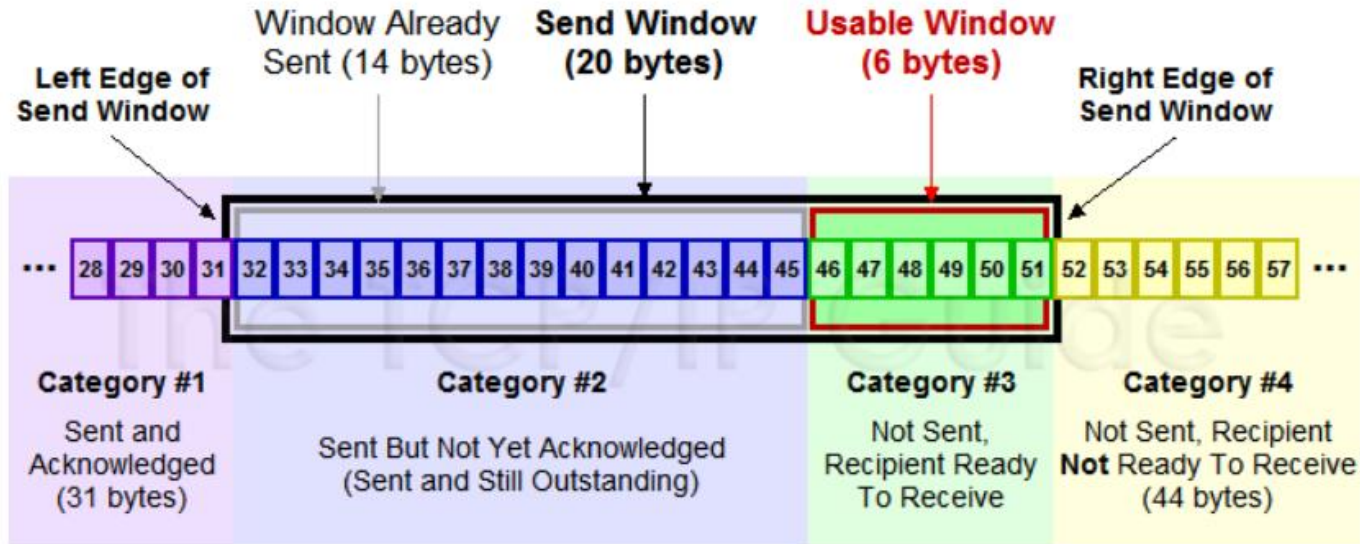
Length: 2

✓ [Timestamps]

0000	00 0c 42 bd b8 f5 90 61 ae 5c 69 58 08 00 45 00	..B....a..iX..E.
0010	00 34 53 6b 40 00 40 06 58 97 c0 a8 58 95 80 77	.4Sk@.@.X...X..w
0020	f5 0c d6 d4 00 50 d1 8b e9 fe 00 00 00 00 80 02	...M.P... ..
0030	fa f0 53 35 00 00 02 04 05 b4 01 03 03 08 01 01	..S5.....
0040	04 02	

mss

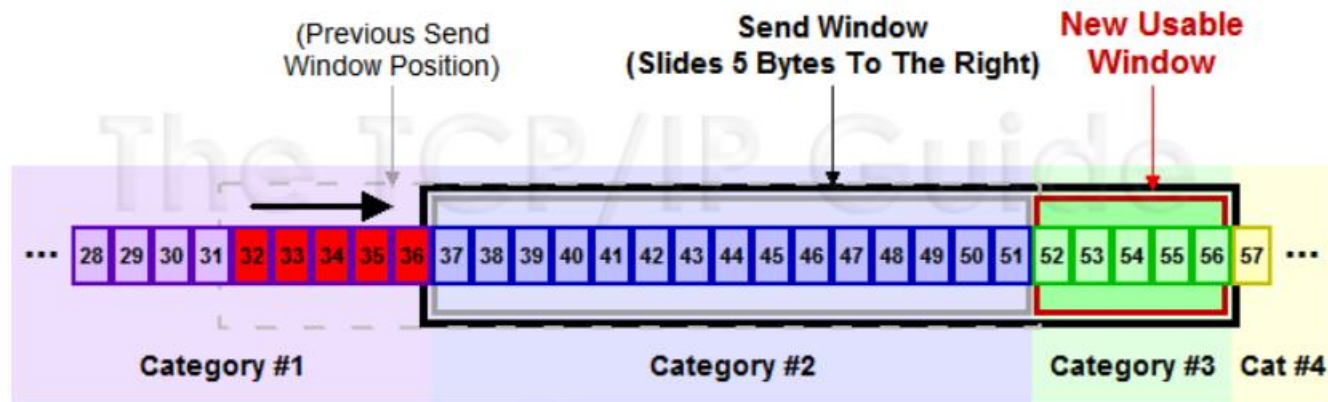
Part B.3 Sliding window system(1)



The **send window** is the key to the entire TCP sliding window system: it represents the maximum number of unacknowledged bytes a device is allowed to have outstanding at once.

The **usable window** is the number of bytes that the sender is still allowed to send at any point in time; it is equal to the size of the send window less the number of unacknowledged bytes already transmitted.

Sliding window system(2)



When the sending device **receives new acknowledgment**, it will be able to transfer some of the bytes from Category #2 to Category #1, since they have now been acknowledged. When it does so, something interesting will happen. Since five bytes have been acknowledged, and the window size didn't change, the sender is allowed to send five more bytes. In effect, the window shifts, or *slides*, over to the right in the timeline.

At the same time five bytes move from Category #2 to Category #1, five bytes move from Category #4 to Category #3, **creating a new usable window for subsequent transmission**.

ACK number, Sequence number, len

No.	Time	Source	Destination	Protocol	Info
94	8.574280	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=65701 Ack=333 Win=30336 Len=1460
95	8.576343	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=67161 Ack=333 Win=30336 Len=1460
96	8.576345	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=68621 Ack=333 Win=30336 Len=1460
✓ 97	8.576345	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=70081 Ack=333 Win=30336 Len=1460
98	8.576516	192.168.88.149	gaia.cs.umass....	TCP	54861 → http(80) [ACK] Seq=333 Ack=71541 Win=65536 Len=0
<					

Source Port: 54861 (54861)

Destination Port: http (80)

[Stream index: 0]

[TCP Segment Len: 0]

Sequence number: 333 (relative sequence number)

[Next sequence number: 333 (relative sequence number)]

Acknowledgment number: 71541 (relative ack number)

0101 = Header Length: 20 bytes (5)

✓ Flags: 0x010 (ACK)

000. = Reserved: Not set

...0 = Nonce: Not set

.... 0... = Congestion Window Reduced (CWR): Not set

.... .0.. = ECN-Echo: Not set

.... ..0. = Urgent: Not set

.... .. 1 = Acknowledgment: Set

.... 0... = Push: Not set

....0.. = Reset: Not set

....0. = Syn: Not set

....0 = Fin: Not set

[TCP Flags:A.....]

ack_num (71541) =
seq (70081) + len (1460)

Changes of window

While the size of usable window turn to be 0, it means the sender will not send any segment at this moment. Wireshark mark the segment with “[Tcp Window Full]”

$$\text{Seq (135781)} + \text{len (1280)} - \text{ack (135781)} = \text{win (1280)}$$

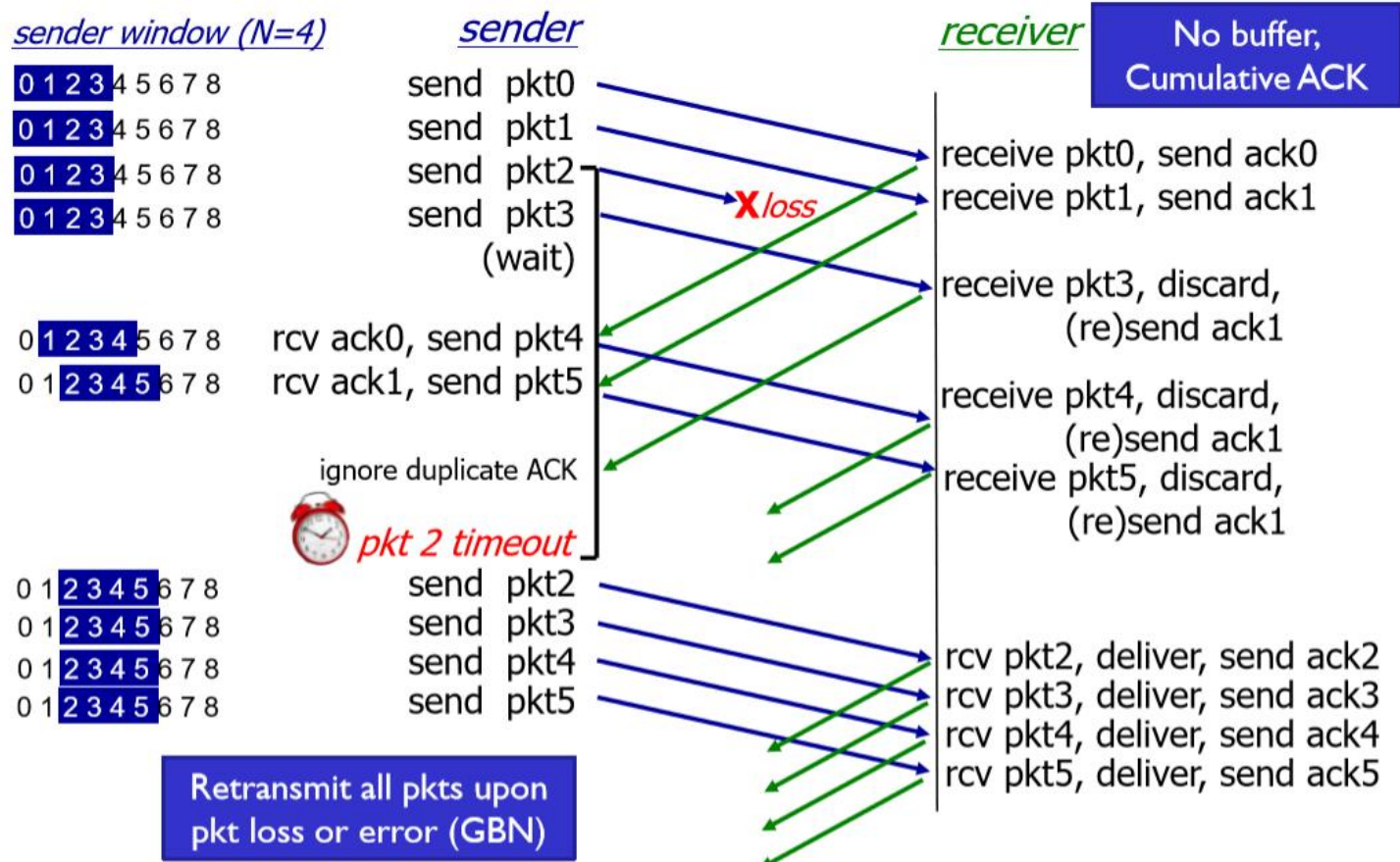
No.	Time	Source	Destination	Protocol	Info
307	19.117577	LAPTOP-RITC8...	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=135781 Win=1280 Len=0
309	20.576025	gaia.cs.umass...	LAPTOP-RITC8FU...	TCP	[TCP Window Full] http(80) → 54861 [PSH, ACK] Seq=135781 Ack=333 Win=30336 Len=1280 [TCP...

While the rcv window turn to be zero, sender will stop to send packet, it will send “[TCP Keep-Alive]” to keep the TCP connection, waiting for the changing of rcv window.

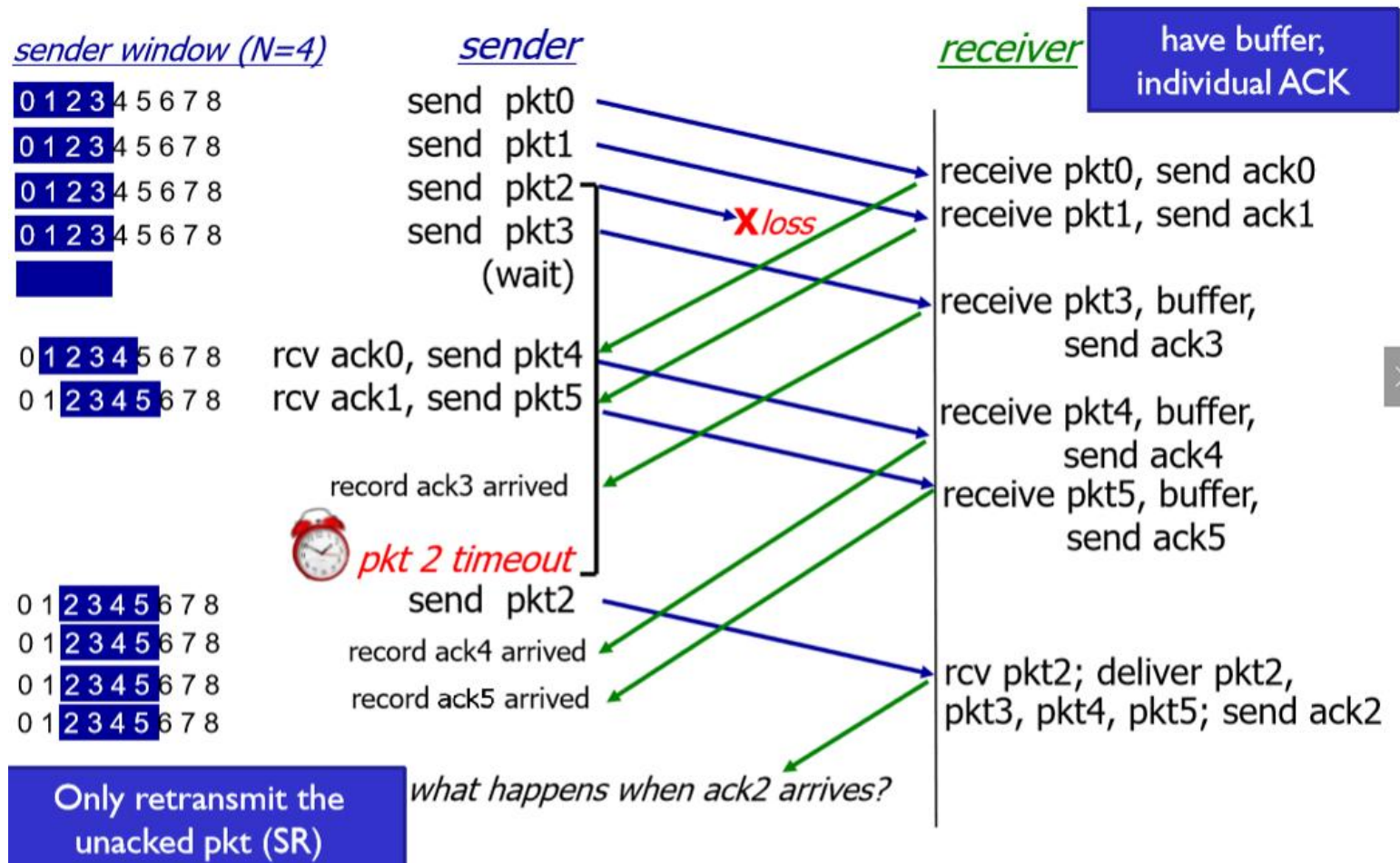
175	19.366403	192.168.88.149	gaia.cs.umass.edu	TCP	[TCP ZeroWindow] 54861 → http(80) [ACK] Seq=333 Ack=137061 Win=0 Len=0
176	20.862900	gaia.cs.umass.edu	192.168.88.149	TCP	[TCP Keep-Alive] http(80) → 54861 [ACK] Seq=137060 Ack=333 Win=30336 Len=0
177	20.862992	192.168.88.149	gaia.cs.umass.edu	TCP	[TCP ZeroWindow] 54861 → http(80) [ACK] Seq=333 Ack=137061 Win=0 Len=0
178	26.701220	gaia.cs.umass.edu	192.168.88.149	TCP	[TCP Keep-Alive] http(80) → 54861 [ACK] Seq=137060 Ack=333 Win=30336 Len=0
179	26.701357	192.168.88.149	gaia.cs.umass.edu	TCP	[TCP ZeroWindow] 54861 → http(80) [ACK] Seq=333 Ack=137061 Win=0 Len=0
180	31.323807	192.168.88.149	gaia.cs.umass.edu	TCP	[TCP Window Update] 54861 → http(80) [ACK] Seq=333 Ack=137061 Win=65536 Len=0

Part B.4 Retransmission

GBN(Go Back N)



SR (Selective Repeat)



TCP SACK

- A Selective Acknowledgment (**SACK**) mechanism, combined with a **selective repeat retransmission policy**, can help the sender **retransmit only the missing data segments**.
 - The receiving TCP **sends back SACK packets to the sender** informing the sender of data that has been received.
 - With selective acknowledgments, the data receiver can inform the sender about all segments that have arrived successfully, so the sender need **retransmit only the segments that have actually been lost**.
- The selective acknowledgment extension uses two TCP options:
 - **SACK-permitted** option
 - **SACK** option

<https://tools.ietf.org/html/rfc2018>

SACK-permitted option

- "SACK-permitted", which may be sent in a SYN segment to indicate that the SACK option can be used once the connection is established.

```
tcp.stream eq 0
No.    Time           Source            Destination      Protocol    Info
1      2.857289    192.168.88.149    128.119.245.12   TCP        60040 → 80 [SYN] Seq=
33 3.208995     128.119.245.12    192.168.88.149   TCP        80 → 60040 [SYN, ACK]
> Frame 8: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
> Ethernet II, Src: IntelCor_5c:69:58 (90:61:ae:5c:69:58), Dst: Routerbo_bd:b8:f5 (00:
> Internet Protocol Version 4, Src: 192.168.88.149, Dst: 128.119.245.12
√ Transmission Control Protocol, Src Port: 60040, Dst Port: 80, Seq: 0, Len: 0
  Source Port: 60040
  Destination Port: 80
  [Stream index: 6]
  [TCP Segment Len: 0]
  Sequence number: 0 (relative sequence number)
  [Next sequence number: 0 (relative sequence number)]
  Acknowledgment number: 0
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x002 [SYN]
  Window size value: 65535
  [Calculated window size: 65535]
  Checksum: 0xd7f5 [unverified]
  [Checksum Status: Unverified]
  Urgent pointer: 0
  √ Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Op
    > TCP Option - Maximum segment size: 1460 bytes
    > TCP Option - No-Operation (NOP)
    > TCP Option - Window scale: 8 (multiply by 256)
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    √ TCP Option - SACK permitted
      Kind: SACK Permitted (4)
      Length: 2
```

TCP Sack-Permitted Option:

Kind: 4

```
+-----+
| Kind=4 | Length=2|
+-----+
```

```
8 2.857289    192.168.88.149    128.119.245.12   TCP        60040 → 80 [SYN] Seq=0 Win=65535 Len=
33 3.208995     128.119.245.12    192.168.88.149   TCP        80 → 60040 [SYN, ACK] Seq=0 Ack=1 Win=
> Frame 33: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
> Ethernet II, Src: Routerbo_bd:b8:f5 (00:0c:42:bd:b8:f5), Dst: IntelCor_5c:69:58 (90:61:ae:5c:69:58)
> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.88.149
√ Transmission Control Protocol, Src Port: 80, Dst Port: 60040, Seq: 0, Ack: 1, Len: 0
  Source Port: 80
  Destination Port: 60040
  [Stream index: 6]
  [TCP Segment Len: 0]
  Sequence number: 0 (relative sequence number)
  [Next sequence number: 0 (relative sequence number)]
  Acknowledgment number: 1 (relative ack number)
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x012 [SYN, ACK]
  Window size value: 29200
  [Calculated window size: 29200]
  Checksum: 0x1a9f [unverified]
  [Checksum Status: Unverified]
  Urgent pointer: 0
  √ Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), Sack permitted,
    > TCP Option - Maximum segment size: 1460 bytes
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    √ TCP Option - SACK permitted
      Kind: SACK Permitted (4)
      Length: 2
```

Wireshark tips: TCP.option_kind==4

SACK option(1)

The **SACK option** is to be sent by a data receiver to inform the data sender of non-contiguous blocks of data that have been received and queued.

The data receiver awaits the receipt of data (perhaps by means of retransmissions) to fill the gaps in sequence space between received blocks.

When missing segments are received, **the data receiver acknowledges the data normally** by advancing the left window edge in the Acknowledgement Number Field of the TCP header.

The SACK option does not change the meaning of the Acknowledgement Number field.

This option contains a list of **some of the blocks of contiguous sequence space occupied by data that has been received and queued within the window.**

Each contiguous block of data queued at the data receiver is defined in the SACK option by **two 32-bit unsigned integers** in network byte order:

- **Left Edge** of Block This is the first sequence number of this block.
- **Right Edge** of Block This is the sequence number immediately following the last sequence number of this block.

Each block represents received bytes of data that are **contiguous and isolated**; that is, **the bytes just below the block, (Left Edge of Block - 1), and just above the block, (Right Edge of Block), have NOT been received.**

SACK option(2)

- SACK option, which may be sent over an established connection once permission has been given by SACK-permitted.

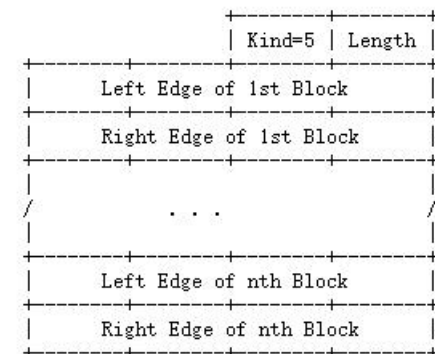
```
No. Time Source Destination Protocol Info
10 1.982570 192.168.88.149 128.119.245.12 TCP 54861 → 80 [ACK] Seq=333 Ack=2921 Wi
11 1.982648 192.168.88.149 128.119.245.12 TCP [TCP Dup ACK 10#1] 54861 → 80 [ACK]

> Frame 10: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
> Ethernet II, Src: IntelCor_5c:69:58 (90:61:ae:5c:69:58), Dst: Routerbo_bd:b8:f5 (00:0c:42:bd:b8:f5)
> Internet Protocol Version 4, Src: 192.168.88.149, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 54861, Dst Port: 80, Seq: 333, Ack: 2921, Len: 0
  Source Port: 54861
  Destination Port: 80
  [Stream index: 0]
  [TCP Segment Len: 0]
  Sequence number: 333 (relative sequence number)
  [Next sequence number: 333 (relative sequence number)]
  Acknowledgment number: 2921 (relative ack number)
  1000 .... = Header Length: 32 bytes (8)
> Flags: 0x010 (ACK)
  Window size value: 256
  [Calculated window size: 65536]
  [Window size scaling factor: 256]
  Checksum: 0xb542 [unverified]
  [Checksum Status: Unverified]
  Urgent pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), SACK
    > TCP Option - No-Operation (NOP)
    > TCP Option - No-Operation (NOP)
    > TCP Option - SACK 4381-5841
      Kind: SACK (5)
      Length: 10
      left edge = 4381 (relative)
      right edge = 5841 (relative)
      [TCP SACK Count: 1]
```

TCP SACK Option:

Kind: 5

Length: Variable



Wireshark tips: TCP.option_kind==5

SACK option(3)

No.	Time	Source	Destination	Protocol	Info
199	46.985513	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=151841 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]
200	46.985600	192.168.88.149	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=153301 Win=65536 Len=0
201	47.595142	gaia.cs.umass...	192.168.88.149	TCP	[TCP Previous segment not captured] http(80) → 54861 [ACK] Seq=156221 Ack=333 Win=30336 Len=1460 [TCP ...
202	47.595144	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=157681 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]
203	47.595274	192.168.88.149	gaia.cs.umass...	TCP	[TCP Dup ACK 200#1] 54861 → http(80) [ACK] Seq=333 Ack=153301 Win=65536 Len=0 SLE=156221 SRE=157681
204	47.595443	192.168.88.149	gaia.cs.umass...	TCP	[TCP Dup ACK 200#2] 54861 → http(80) [ACK] Seq=333 Ack=153301 Win=65536 Len=0 SLE=156221 SRE=159141
205	48.207253	gaia.cs.umass...	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=153301 Ack=333 Win=30336 Len=1460
206	48.207367	192.168.88.149	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=154761 Win=65536 Len=0 SLE=156221 SRE=159141
207	49.742628	gaia.cs.umass...	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=154761 Ack=333 Win=30336 Len=1460
208	49.742765	192.168.88.149	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=159141 Win=65536 Len=0
209	50.363845	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=159141 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]

#203 and #204 are SACK

#203 tells that 156221~157681 are **contiguous and isolated**

#204 tells that 156221~159141 are **contiguous and isolated**

#200 tells that the block before 153301 are acked

So

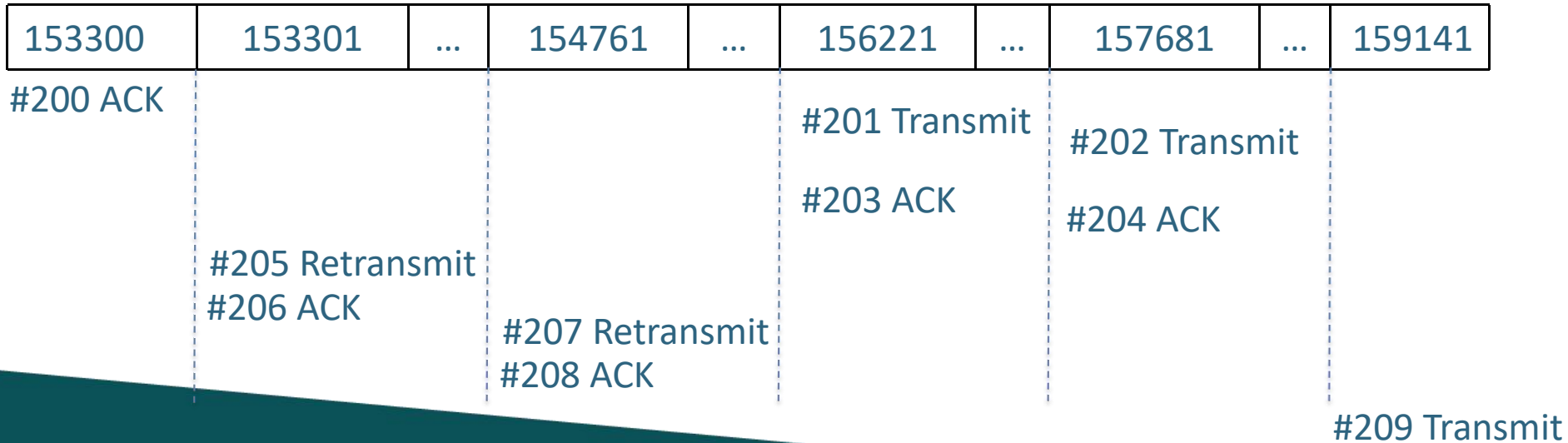
#205 retransmit 153301 ~ 153301+ 1460 -1 ,#206 ack it with 154761

#207 retransmit 154761 ~ 154761+1460 -1

#208 ack it with 159141 (for 157681~159140 are **contiguous but NOT isolated**)

SACK option(4)

No.	Time	Source	Destination	Protocol	Info
199	46.985513	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=151841 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]
200	46.985600	192.168.88.149	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=153301 Win=65536 Len=0
201	47.595142	gaia.cs.umass...	192.168.88.149	TCP	[TCP Previous segment not captured] http(80) → 54861 [ACK] Seq=156221 Ack=333 Win=30336 Len=1460 [TCP ...
202	47.595144	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=157681 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]
203	47.595274	192.168.88.149	gaia.cs.umass...	TCP	[TCP Dup ACK 200#1] 54861 → http(80) [ACK] Seq=333 Ack=153301 Win=65536 Len=0 SLE=156221 SRE=157681
204	47.595443	192.168.88.149	gaia.cs.umass...	TCP	[TCP Dup ACK 200#2] 54861 → http(80) [ACK] Seq=333 Ack=153301 Win=65536 Len=0 SLE=156221 SRE=159141
205	48.207253	gaia.cs.umass...	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=153301 Ack=333 Win=30336 Len=1460
206	48.207367	192.168.88.149	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=154761 Win=65536 Len=0 SLE=156221 SRE=159141
207	49.742628	gaia.cs.umass...	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=154761 Ack=333 Win=30336 Len=1460
208	49.742765	192.168.88.149	gaia.cs.umass...	TCP	54861 → http(80) [ACK] Seq=333 Ack=159141 Win=65536 Len=0
209	50.363845	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=159141 Ack=333 Win=30336 Len=1460 [TCP segment of a reassembled PDU]



Retransmission(1)

No.	Time	Source	Destination	Protocol	Info
202	47.595144	gaia.cs.umass...	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=157681 Ack=333 Win=30336 Len=146
203	47.595274	192.168.88.149	gaia.cs.umass...	TCP	[TCP Dup ACK 200#1] 54861 → http(80) [ACK] Seq=333 Ack=1533
204	47.595443	192.168.88.149	gaia.cs.umass...	TCP	[TCP Dup ACK 200#2] 54861 → http(80) [ACK] Seq=333 Ack=1533
205	48.207253	gaia.cs.umass...	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=153301 Ack=

Sequence number: 153301 (relative sequence number)
[Next sequence number: 154761 (relative sequence number)]
Acknowledgment number: 333 (relative ack number)
0101 = Header Length: 20 bytes (5)
> Flags: 0x010 (ACK)
Window size value: 237
[Calculated window size: 30336]
[Window size scaling factor: 128]
Checksum: 0x3487 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0

√ [SEQ/ACK analysis]
[iRTT: 0.450320000 seconds]
[Bytes in flight: 5840]
[Bytes sent since last PSH flag: 16060]

√ [TCP Analysis Flags]
√ [Expert Info (Note/Sequence): This frame is a (suspected) retransmission]
[This frame is a (suspected) retransmission]
[Severity level: Note]
[Group: Sequence]
[The RTO for this segment was: 0.612109000 seconds]
[RTO based on delta from frame: 202]

While RTO timeout , retransmission is triggered

Retransmission(2)

No.	Time	Source	Destination	Protocol	Info
202	47.595144	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=157681 Ack=333 Win=30336 Len=1466
203	47.595274	192.168.88.149	gaia.cs.umass....	TCP	[TCP Dup ACK 200#1] 54861 → http(80) [ACK] Seq=333 Ack=15336
204	47.595443	192.168.88.149	gaia.cs.umass....	TCP	[TCP Dup ACK 200#2] 54861 → http(80) [ACK] Seq=333 Ack=15336
205	48.207253	gaia.cs.umass....	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=153301 Ack=3
206	48.207367	192.168.88.149	gaia.cs.umass....	TCP	54861 → http(80) [ACK] Seq=333 Ack=154761 Win=65536 Len=0 SL
207	49.742628	gaia.cs.umass....	192.168.88.149	TCP	[TCP Retransmission] http(80) → 54861 [ACK] Seq=154761 Ack=3

Sequence number: 154761 (relative sequence number)
[Next sequence number: 156221 (relative sequence number)]
Acknowledgment number: 333 (relative ack number)
0101 = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window size value: 237
[Calculated window size: 30336]
[Window size scaling factor: 128]
Checksum: 0x595f [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0

✓ [SEQ/ACK analysis]
[iRTT: 0.450320000 seconds]
[Bytes in flight: 4380]
[Bytes sent since last PSH flag: 17520]

✓ [TCP Analysis Flags]
✓ [Expert Info (Note/Sequence): This frame is a (suspected) retransmission]
[This frame is a (suspected) retransmission]
[Severity level: Note]
[Group: Sequence]
[The RTO for this segment was: 2.147484000 seconds]
[RTO based on delta from frame: 202]

Fast retransmission

- TCP may generate an immediate acknowledgment (a duplicate ACK) when an out-of-order segment is received. This duplicate ACK should not be delayed. The purpose of this duplicate ACK is to let the other end know that a segment was received out of order, and to tell it what sequence number is expected.
- Since TCP does not know whether a duplicate ACK is caused by a lost segment or just a reordering of segments, it waits for a small number of duplicate ACKs to be received.
 - It is assumed that if there is just a reordering of the segments, there will be only one or two duplicate ACKs before the reordered segment is processed, which will then generate a new ACK.
 - If three or more duplicate ACKs are received in a row, it is a strong indication that a segment has been lost.
- TCP then performs a retransmission of what appears to be the missing segment, without waiting for a retransmission timer to expire.

115	10.757197	192.168.88.149	gaia.cs.umass....	TCP	TCP Dup ACK 113#1	54861 → http(80) [ACK] Seq=333	Ack=87601	Win=49408	Len=0	SLE=90521	SRE=91981
116	10.758693	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=91981	Ack=333	Win=30336	Len=1460	[TCP segment of a reassembled PDU]		
117	10.758765	192.168.88.149	gaia.cs.umass....	TCP	TCP Dup ACK 113#2	54861 → http(80) [ACK] Seq=333	Ack=87601	Win=49408	Len=0	SLE=90521	SRE=93441
118	11.340240	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=93441	Ack=333	Win=30336	Len=1460	[TCP segment of a reassembled PDU]		
119	11.340311	192.168.88.149	gaia.cs.umass....	TCP	TCP Dup ACK 113#3	54861 → http(80) [ACK] Seq=333	Ack=87601	Win=49408	Len=0	SLE=90521	SRE=94901
120	11.341761	gaia.cs.umass....	192.168.88.149	TCP	http(80) → 54861 [ACK] Seq=94901	Ack=333	Win=30336	Len=1460	[TCP segment of a reassembled PDU]		
121	11.341762	gaia.cs.umass....	192.168.88.149	TCP	TCP Fast Retransmission	http(80) → 54861 [ACK] Seq=87601	Ack=333	Win=30336	Len=1460	[TCP segment of	

<https://tools.ietf.org/html/rfc2001>

Practice 7.1

7.1 Select one UDP packet from your trace. From this packet,

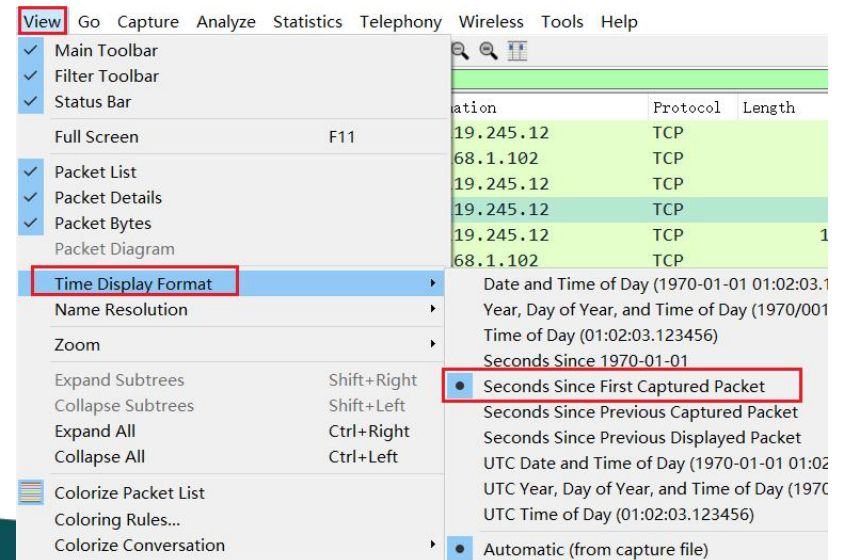
- determine
 - Q1) how many fields are there in the UDP header.
 - Q2) what are the name and value of each field in the UDP header.
 - Q3) the length (in bytes) of each fields in the UDP header.
 - Q4) What is the maximum number of bytes of a UDP packet ? (Hint: the answer can be determined by your answer to Q3 above)
 - Q5) What is the largest possible destination port number? (Hint: same as the hint in Q4 above.)
 - Q6) What is the protocol ID for UDP in IP protocol?(Give your answer in both hexadecimal and decimal notation.)

Practice 7.2

- Finish the question 1, 2, 4, 5, 6, 11, 12 of Wireshark_TCP_v8.0.pdf
- Tips: To calculate the throughput for the TCP connection, we can use following formula:

$$\text{Throughput (Bytes/s)} = \text{Total data(Bytes)} / \text{Total time(s)}$$

- Tips: It's convenient to use “seconds since first captured packet” time format.

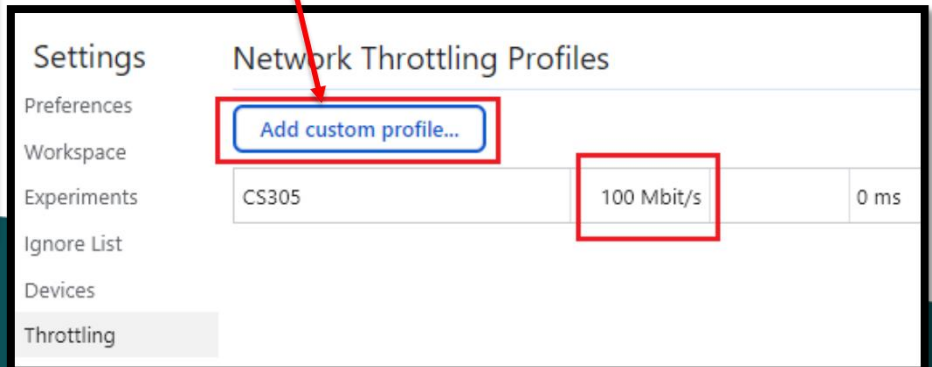
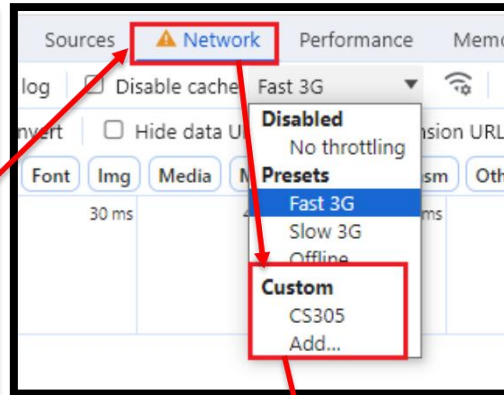
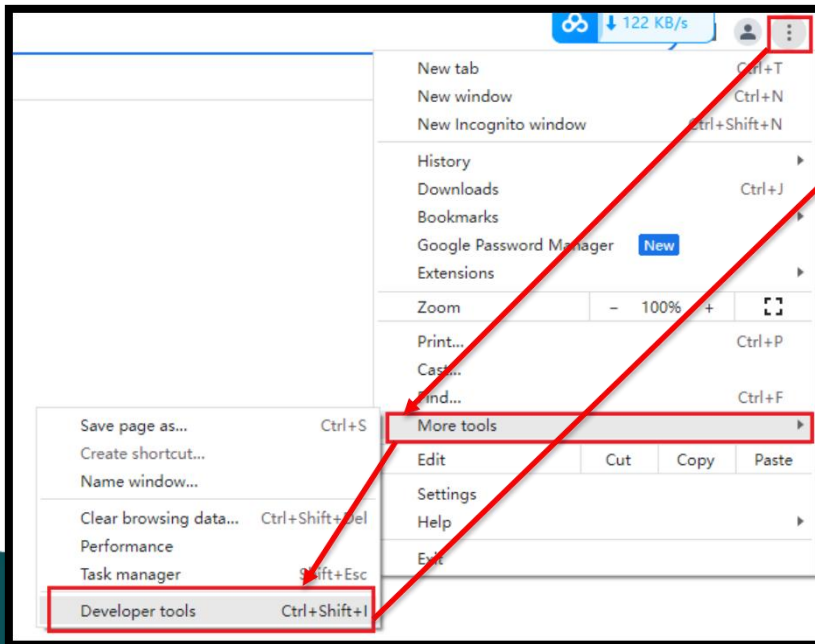


Practice 7.3

- Using Wireshark to capture and analysis the TCP stream.
- Invoke a HTTP request to get <http://gaia.cs.umass.edu/wiresharklabs/alice.txt>
- Analysis the TCP stream
 - Any duplicate ack, what's the possible reason?
 - Any TCP segment with sack permit option and sack option
 - select a tcp package which contained a sack option, find the segment ranges which is acked in this sack option
 - Any TCP retransmission? Is it retransmission or fast retransmission?
 - Any window size 0 segment, what does it mean? If the window size is 0, what would happened next on this tcp connection?
 - Any TCP window full segment, what does it mean?

Tips (1)

- How to cause congestion on your network.
 - You can use some tools, such as clumsy-0.2-win64, or Burp Suite, ...
 - You can make your browser work in a slow mode.



Tips (2)

- How to get all the TCP segments of the TCP stream related to a http session in Wireshark:
 - 1st step: Find a HTTP packet in the HTTP session
 - 2nd step: right click the packet which will invoke a shortcut menu, then choose “follow ->TCP” in the shortcut menu
- How to find a special segment in a TCP stream
 - Using view filter in Wireshark
 - 1st step: if a TCP stream is filtered, then a description such as “TCP.stream eq xx” (xx here is the id of this TCP stream) could be found in the view filter
 - 2nd step: in the same view filter, make a new filter rule description along with the original one, then press “Enter” key in your keyboard to make the new filter run
 - such as :
TCP.stream eq 1 is the original one, to find a TCP zero window in this TCP stream,
TCP.stream eq 1 && TCP.window_size_value==0 is the new view filter description

Tips (3)

- Analysis -> expert info
- Statistic->TCP stream graphs

Wireshark · Expert Information · download_tcp_flow_1.pcapng

Severity	Summary	Group	Protocol	Count
> Warning	TCP Zero Window segment	Sequence	TCP	3
> Warning	TCP window specified by the receiver is now co...	Sequence	TCP	1
> Warning	This frame is a (suspected) out-of-order segment	Sequence	TCP	24
> Warning	Previous segment(s) not captured (common at c...	Sequence	TCP	45
> Note	TCP keep-alive segment	Sequence	TCP	2
> Note	This frame is a (suspected) fast retransmission	Sequence	TCP	5
> Note	This frame is a (suspected) retransmission	Sequence	TCP	48
> Note	Duplicate ACK (#1)	Sequence	TCP	137
> Chat	Connection finish (FIN)	Sequence	TCP	2
> Chat	TCP window update	Sequence	TCP	1
> Chat	GET /wireshark-labs/wireshark-traces.zip HTTP/1....	Sequence	HTTP	1
> Chat	Connection establish acknowledge (SYN+ACK): s...	Sequence	TCP	1
> Chat	Connection establish request (SYN): server port 80	Sequence	TCP	1

